

Lab 6 - Math 58B: error rates

due Tuesday March 3, 2026

your name here

```
library(praise)
```

Lab Goals

By assessing our own comfort with randomness, we investigate 0.05. Additionally, we analyze data without a clear decision as to whether or not the null hypothesis should be rejected.

- why 0.05?
- what does a p-value really mean?
- how / why is there (is there not) always a single result for a research questions?

Getting started

Load packages

In this lab we will continue to use **infer** syntax and the `xpnorm()` function which is in the **mosaic** package.

Let's load the packages.

```
library(tidyverse) # ggplot and wrangling
library(mosaic)    # xpnorm and xqnorm
library(infer)     # simulation inference code
```

Preliminary personal level of discernibility

Follow the link here and click through the scatterplot images: <https://www.openintro.org/book/stat/why05/>

- (a) What is your personal level of discernibility?
- (b) When you hear new information, do you consider yourself on the skeptical side or on the believing side? (There is no right answer!)

The data

(The data is only for Q1. The other questions do not require data.)

Researchers have conjectured that the use of the word “forbid” is more off-putting than the word “allow” (in affecting people’s responses to survey questions). In particular, the suggestion is that people do not like to “forbid” anything. Students in an introductory statistics class were randomly assigned to answer one of the following questions:

- Should your college allow speeches on campus that might incite violence?
- Should your college forbid speeches on campus that might incite violence?

Of the 14 students who received the first question, 8 responded yes. Of the 15 students who received the second question, 13 said no. Think carefully about the response variable. It should *not* be coded as “yes” and “no” as answered on the questionnaire.

What do the data look like?

Using the information above, create a 2x2 table (with your pencil) describing the dataset. What are your two variables? How many people are in each group? Again, do not code with “yes” and “no”, use descriptive words.

Hypothesis test

In order to formally test the researchers’ conjecture (that the words can be off-putting), a null sampling distribution must be created (in order to compare the observed data against). You will create the null sampling distribution in two ways:

1. Using the **infer** syntax to do a randomization test.
2. Using the Central Limit Theorem to do a Z test.

The Central Limit Theorem: the sample average (e.g., sample mean, sample proportion, etc.) in repeated random samples taken from a population will be normally distributed if the sample size is large enough.

Note that for the two proportion case, if the sample sizes are large enough the CLT says:

$$\hat{p}_1 - \hat{p}_2 \sim N\left(p_1 - p_2, \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}\right)$$

It isn't a hard and fast rule, but generally the CLT approximation is good if there are at least 10 (or possibly 5) successes and at least 10 (possibly 5) failures in each group.

To Turn In

Q1. Speeches and Violence

In this first question, you will analyze the data described above using some old ideas and some new ideas.

To create a dataset, use the `rep()` function to replicate an appropriate number of times. Use the `c()` function to create a column. You should be able to do this on your own. For now, fill in the blanks (and then change to `eval = TRUE`).

```
# first create a data frame with the survey data
decision <- data.frame( ___ = c(rep("___", 14), rep("___", 15)),
                        ___ = c(rep("___", 8), rep("___", 6),
                                rep("___", 13), rep("___", 2)))
```

- Plot the observed data using `geom_bar()` and use `fill = response` to fill the bars in with appropriate colors, where the word **response** represents whatever you called the variable representing how the students responded to the survey.
- Use **infer** to analyze the data using a randomization test. Report the one-sided p-value (you will report the conclusion in words below in part d.).
- Use the following formula to create a Z score for the same hypothesis test (different p-value calculation) as was done with **infer** (this time it is called a Z test). Use R as a calculator to find the relevant Z score, and find the one-sided p-value (you will report the conclusion in words below in part d.).

$$Z \text{ score} = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}}$$

- d. Give a complete conclusion to the data analysis / hypothesis test (that is, conclude what you think is most appropriate). State the null and alternative hypotheses, provide what you believe is the most accurate discernibility result (compare parts b. and c. above), and give a sense of to whom (what population, if any) the results can be applied. (Hint for the conclusion part: the CLT holds for large sample sizes. Our samples aren't huge here.)
- e. (Now the goal is to practice ideas of power. Note that the sample sizes are small enough that the CLT might not hold, but we are going to assume it does for part e.) Let's pretend that you didn't reject H_0 when you applied the mathematical model. You might want to reflect on whether or not you had enough power for the test.
- To do the problem, we'll need to approximate the sample proportions in the standard error. For simplicity (and because it doesn't change the SE very much), let's assume that

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} = \sqrt{\frac{0.5(1 - 0.5)}{14} + \frac{0.5(1 - 0.5)}{15}} = 0.186$$

- Your PI (Principal Investigator = research boss) tells you that she had originally expected the difference in sample proportions to be -0.25.
- You know that you would have rejected H_0 if your Z score was less than -1.645 (if your level of discernibility is 0.05).
 - i. What value of $\hat{p}_1 - \hat{p}_2$ would make you reject H_0 (assume $SE(\hat{p}_1 - \hat{p}_2) = 0.186$)?
 - ii. Estimate the power of the test when the true difference is $p_1 - p_2 = -0.25$ (assume $SE(\hat{p}_1 - \hat{p}_2) = 0.186$)?

Q2. p-values

Read the ASA's statement on p-values:

<http://www.tandfonline.com/doi/pdf/10.1080/00031305.2016.1154108>

Choose **two different** principles (pg 131-132) and explain each (separately) as if to a peer in a science class who is making conclusions about a recent study. Explain in your own words.

Recall: significance = discernibility

Q3. p-values, take two

Why use p-values at all? That is, what is benefit of having a p-value (as opposed to simply descriptive statistics or graphs of the data)?

If you are still curious about the ideas in this lab try investigating some of the following (not part of the assignment):

None of the queries below are part of the lab. I offer them here for people who are intrigued by the ideas we've covered and want to know more. Indeed, the article linked below (which has been cited 9000+ times and viewed more than 3 million times) has a provocative title (and is well written).

Q5. Read Ioannidis (2005), "Why Most Published Research Findings are False" <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.0020124>

- (a) Consider table 1. Suppose that the level of discernibility is taken to be 0.05 and the power is 0.8. Also, set R (the number of true to not true relationships) to be 2 (for every 3 experiments, one is null). What percent of research findings (i.e., "discernible" findings) are actually true (i.e., H_A is true)? [hint: for ease of calculation, you can set c to be something like 10,000.]
- (b) Consider table 1. Suppose that the level of discernibility is taken to be 0.05 and the power is 0.3. Also, set R (the number of true to not true relationships) to be 0.1 (for every 11 experiments, 10 are null). What percent of research findings (i.e., "discernible" findings) are actually true (i.e., H_A is true)? [hint: for ease of calculation, you can set c to be something like 10,000.]

Q6. In the Dance of the p-values (<https://www.youtube.com/watch?v=5OL1RqHrZQ8>), what is the narrator arguing?

```
praise()
```

```
[1] "You are supreme!"
```